Homework 29: Linear Recurrence Relations – Solving via Characteristic Polynomial

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Introduction. One of the most useful aspects of diagonalization and matrix functions is their ability to swiftly solve problems that do not even seem to be linear algebraic in nature at first glance. We have already seen a hint of how this works in *homework 8* where we developed a closed-form formula for the Fibonacci numbers. In this homework, we extend this idea to a larger family of sequences of numbers $x_1, x_2, x_3, \ldots$ that are defined in a similar linear way.

A linear recurrence relation is a formula that represents a term x_n in a sequence of numbers as a linear combination of the immediately previous terms in the sequence. That is, it is a formula for which there exist scalars $a_0, a_1, \ldots, a_{k-1}$ so that

$$x_n = a_{k-1}x_{n-1} + a_{k-2}x_{n-2} + \cdots + a_0 x_{n-k}, \qquad \forall, \qquad n \geq k\,. \tag{1}$$

The number of terms in the linear combination is called the degree of the recurrence relation (so the recurrence relation has degree k). In order for the sequence to be completely defined, we also need some initial conditions that specify the first term(s) of the sequence. In particular, for a degree-k recurrence relation we require the first k terms to be specified.

The characteristic polynomial of the degree-k linear recurrence relation

$$x_n = a_{k-1}x_{n-1} + a_{k-2}x_{n-2} + \cdots + a_0 x_{n-k}\,, \tag{2}$$

is the degree-k polynomial

$$p(\lambda) = \lambda^k - a_{k-1}\lambda^{k-1} - a_{k-2}\lambda^{k-2} - \cdots - a_1\lambda - a_0\,. \tag{3}$$

Theorem 1. ***Solving Linear Recurrence Relations.*** Suppose $x_0, x_1, x_2, \ldots$ is a sequence satisfying a linear recurrence relation whose characteristic polynomial has distinct roots $\lambda_0, \lambda_1, \ldots, \lambda_{k-1}$. Let $c_0, c_1, \ldots, c_{k-1}$ be the (necessarily unique) scalars such that

$$x_n = c_0\lambda_0^n + c_1\lambda_1^n + c_2\lambda_2^n + \cdots + c_{k-1}\lambda_{k-1}^n, \qquad when\ 0 \leq n < k\,. \tag{T1}$$

Then the formula (T1) holds for all $n \geq 0$.

Linear System Involving Coefficients. ***Finding Unknown Scalars $c_i's$.*** The fact that we can find scalars $c_0, c_1, \ldots, c_{k-1}$ satisfying Equation (T1) can be made clearer by writing it out as a linear system. Explicitly, this linear system has the

$$\begin{bmatrix} 1 & 1 & \cdots & 1 \\ \lambda_0 & \lambda_1 & \cdots & \lambda_{k-1} \\ \lambda_0^2 & \lambda_1^2 & \cdots & \lambda_{k-1}^2 \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_0^{k-1} & \lambda_1^{k-1} & \cdots & \lambda_{k-1}^{k-1} \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ c_2 \\ \vdots \\ c_{k-1} \end{bmatrix} = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ \vdots \\ x_{k-1} \end{bmatrix}.$$

Remark. For example, plugging $n = 0$ into Equation (T1) gives the first equation in this linear system, plugging in $n = 1$ gives the second equation, and so on.

Problem 1. Find the Fibonacci recurrence $F_0 = 0, F_1 = 1$, and $F_n = F_{n-1} + F_{n-2}$, for $n \geq 2$.

Note. Formula *"found"* is a remarkable formula, because it is defined in terms of the irrational number $\sqrt{5}$ yet the Fibonacci numbers are all integers! Try plugging in a few values for n to see how the $\sqrt{5}$ terms cancel out to leave the integer values F_n. The formula *"found"* is known as ***Binet's formula***.

The method we have just outlined works for any second order linear recurrence relation whose associated eigenvalues are all distinct. When there is a repeated eigenvalue, the technique must be modified, since the diagonalization method we used may no longer work. The next theorem summarizes both situations.

Theorem 2. Let $x_n = ax_{n-1} + bx_{n-2}$ be a recurrence relation that is satisfied by a sequence (x_n). Let λ_1 and λ_2 be the eigenvalues of associated characteristic equation $\lambda^2 - a\lambda - b = 0$.

a. If $\lambda_1 \neq \lambda_2$, then $x_n = c_1\lambda_1^n + c_2\lambda_2^n$.

b. If $\lambda_1 = \lambda_2 = \lambda$, then $x_n = c_1\lambda^n + c_2 n\lambda^n$.

Problem 2. ***Repeated Eigenvalues.***

Solve the recurrence relation $x_0 = 1, x_1 = 6$, and $x_n = 6x_{n-1} - 9x_{n-2}$ for $n \geq 2$.

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Good Luck